

Secure File Sharing Application

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled "SECURE FILE SHARING APPLICATION" is the bonafide work of SKANDAN KAMAL (230701322), SIDHARTH SUBRAMONIAN (230701315), JAYAN A (230701128), and SIVARAMAKRISHNAN R (230701319) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

"This project addresses the following Sustainable Development Goal(s): Goal 8 – Decent Work and Economic Growth; Goal 9 – Industry, Innovation and Infrastructure; Goal 16 – Peace, Justice and Strong Institutions."

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ABSTRACT

Modern file-sharing applications often depend on internet connectivity, cloud infrastructure, or platform-specific ecosystems. While solutions such as AirDrop enable local transfers, securely exchanging sensitive files between untrusted devices requires reliable authentication and strong cryptographic protection. This is especially important in environments where internet access is unavailable, unreliable, or restricted, such as disaster-response operations, remote locations, hospitals, defense environments, and air-gapped networks.

This project presents a Secure File Sharing Application supporting offline peer-to-peer file transfers. The system uses the Nearby Connections API, out-of-band pairing verification, and a hybrid post-quantum key exchange using ML-KEM-768 with X25519. A session key is derived using HKDF-SHA256, and files are encrypted using AES-256-GCM, providing confidentiality and integrity without relying on Internet or server infrastructure.

The proposed architecture combines peer-to-peer networking, post-quantum cryptography, and authenticated encryption in a lightweight mobile application. It reduces dependence on Internet and cloud infrastructure, providing confidentiality, integrity, and peer authentication for offline file sharing.

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LIST OF ABBREVIATIONS

AES	Advanced Encryption Standard
API	Application Programming Interface
DFD	Data Flow Diagram
ECC	Elliptic Curve Cryptography
ECDH	Elliptic Curve Diffie-Hellman
GCM	Galois/Counter Mode
HKDF	HMAC-based Key Derivation Function
IoT	Internet of Things
IV	Initialization Vector
KEM	Key Encapsulation Mechanism
ML-DSA	Module-Lattice-Based Digital Signature Algorithm
ML-KEM	Module-Lattice-Based Key Encapsulation Mechanism
MITM	Man-in-the-Middle
MQTT	Message Queuing Telemetry Transport
OOB	Out-of-Band
P2P	Peer-to-Peer
PQ	Post-Quantum
PQC	Post-Quantum Cryptography
PQ/T	Post-Quantum/Traditional
QR	Quick Response
RAM	Random Access Memory
RSA	Rivest-Shamir-Adleman
SDG	Sustainable Development Goal
SHA	Secure Hash Algorithm
SSH	Secure Shell
TLS	Transport Layer Security
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Modern file-sharing applications commonly depend on internet connectivity, cloud infrastructure, or platform-specific ecosystems. While solutions such as AirDrop and other peer-to-peer tools enable local transfers, securely exchanging sensitive files between untrusted devices requires reliable device authentication and strong cryptographic protection.

The challenge is more significant in environments where internet access is unavailable, unreliable, or intentionally restricted — such as disaster-response operations, remote locations, hospitals, defense environments, and air-gapped networks. A compromised or impersonated device could result in sensitive files being intercepted or transferred to the wrong recipient.

This project explores a secure offline peer-to-peer file-sharing system. The proposed system uses out-of-band pairing verification and a hybrid post-quantum key exchange using ML-KEM-768 + X25519 to establish session secrets between nearby devices, without requiring Internet connectivity or server infrastructure.

1.2 MOTIVATION

With the increasing sensitivity of data exchanged between personal and professional devices — medical records, confidential documents, maps, and images — the risk of interception or misdirected transfer has grown substantially. Existing solutions either rely heavily on cloud infrastructure, exposing data to third-party servers, or lack explicit peer identity verification, leaving them vulnerable to man-in-the-middle attacks. Additionally, with the advancement of quantum computing, classical key-exchange algorithms such as RSA and ECC face a long-term threat, motivating the adoption of post-quantum cryptographic primitives even in everyday consumer applications.

1.3 PROJECT OVERVIEW AND OBJECTIVES

Aim: Enable users to securely share files between nearby devices in offline peer-to-peer mode, providing confidentiality, integrity, and peer authentication.

Approach: The system establishes a peer-to-peer connection using the Nearby Connections API, authenticates devices using out-of-band verification, performs a hybrid ML-

KEM-768 + X25519 key exchange, derives a session key using HKDF-SHA256, and encrypts files using AES-256-GCM.

Key objectives:

1. Secure peer authentication through user-verified out-of-band pairing.
2. Hybrid post-quantum key establishment combining ML-KEM-768 with classical X25519 for offline transfers.
3. Confidential and integrity-protected file transfer using authenticated encryption.
4. Minimize dependence on Internet and cloud infrastructure for offline transfers.
5. Secure session teardown by discarding ephemeral private keys and session keys after transfer.
6. Evaluate the feasibility of a lightweight security-focused file-transfer architecture for offline environments.

1.4 SCOPE OF THE PROJECT

Scope — included: Nearby device discovery using the Nearby Connections API, pairing verification, ML-KEM-768 + X25519 key exchange, HKDF-SHA256 key derivation, AES-256-GCM offline file transfer, keychain management, integrity verification, and secure session teardown.

Scope — excluded (future work): Online multi-recipient file sharing via a ciphertext server (planned for Phase II), enterprise/military infrastructure integration, large-scale file distribution, advanced identity and access-management systems, and protection against already-compromised endpoint devices.

1.5 SUSTAINABLE DEVELOPMENT GOALS ADDRESSED

This project addresses the following Sustainable Development Goals:

- Goal 8 — Decent Work and Economic Growth
- Goal 9 — Industry, Innovation and Infrastructure
- Goal 16 — Peace, Justice and Strong Institutions

CHAPTER 2

LITERATURE REVIEW

A wide body of recent research investigates hybrid post-quantum key-exchange mechanisms, the practical deployment of ML-KEM (Kyber) alongside classical algorithms such as X25519 and ECC, and the performance implications of adopting post-quantum cryptography (PQC) on constrained and networked systems. The table below summarizes the key literature reviewed for this project, spanning hybrid key-agreement protocols, authenticated encryption schemes, formal verification of PQ protocols, and performance evaluations across TLS, IoT, 5G, and embedded environments.

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
Kwiatkowski et al. / 2026	Post-Quantum Traditional (PQ/T) Hybrid Key Agreement Mechanisms for TLS 1.3	Combines X25519 ECDHE with ML-KEM-768 for hybrid key establishment.	Supports our ML-KEM-768 + X25519 hybrid key exchange approach.
Prikutse et al. / 2026	A Commitment-Based Hybrid Post-Quantum Cryptographic Model for Multi-File Cloud Storage	Uses AES-256-GCM, X25519 and ML-KEM-768 for encrypted multi-file storage.	Relevant to hybrid post-quantum encryption of stored files; informs planned Phase II work.
Olushola & Meenakshi / 2026	Design and Implementation of an Authenticated Post-Quantum Session Protocol Using ML-KEM, ML-DSA, and AES-256-GCM	Uses ML-KEM, ML-DSA, HKDF and AES-256-GCM for authenticated sessions.	Supports our use of key derivation and authenticated encryption.
Shaw / 2026	quantum-safe: Bridging the Post-Quantum Production Gap with a Hybrid-by-Default Python Cryptography Library	Implements and evaluates X25519 + ML-KEM-768 hybrid cryptography.	Demonstrates practical implementation and performance of hybrid PQ key exchange.
Dubey & Varshney / 2026	Measurement Study of Post-Quantum Readiness of Internet: 2026	Measures deployment and adoption of ML-KEM-768 + X25519 in Internet TLS.	Shows increasing practical adoption of hybrid post-quantum cryptography.
Various Authors / 2026	Micro and Transport Layer Security Benchmarking of Post-Quantum Versus Classical Cryptography	Compares PQ and classical cryptography based on performance and communication overhead.	Highlights the performance considerations of deploying PQ cryptography.
Various Authors / 2026	Performance Evaluation of Post-Quantum Cryptography in IoT: A Case Study on MQTT Over TLS Under Network Constraints	Evaluates PQC performance in constrained and ARM-based environments.	Demonstrates the importance of evaluating PQC on resource-constrained devices.
Shakib, Hafeez & Munir / 2025	AmphiKey: A Dual-Mode Secure Authenticated Key Encapsulation Protocol for Smart Grid	Combines ML-KEM-768 + X25519 with authentication and forward secrecy.	Strongly supports our hybrid key establishment design.
Various Authors / 2025	Optimized and Secure Hybrid Key Exchange Model for Post-Quantum Digital Communications	Combines ECC, Kyber-768 and AES-256-GCM for secure communication.	Demonstrates the use of hybrid cryptography with authenticated encryption.

Author(s) / Year	Title of the Paper	Methodology / Approach	Inference
Various Authors / 2025	Performance Evaluation of Post-Quantum Cryptography Algorithms for Secure Communication in Wireless Networks	Evaluates PQ algorithms combined with symmetric encryption over wireless networks.	Relevant to our secure nearby wireless file transfer scenario.
Chang / 2025	Post-Quantum Key Exchange and Subscriber Identity Encryption in 5G Using ML-KEM (Kyber)	Applies ML-KEM to key exchange and identity protection in 5G networks.	Demonstrates practical application of ML-KEM in secure communication.
Chhetri et al. / 2025	Post-Quantum Cryptography and Quantum-Safe Security: A Comprehensive Survey	Surveys PQ algorithms, hybrid cryptography, migration strategies and deployment challenges.	Provides background for selecting post-quantum cryptographic techniques.
Various Authors / 2025	Enhanced Network Security Protocols for the Quantum Era: Combining Classical and Post-Quantum Cryptography, and Quantum Key Distribution	Proposes hybrid classical and post-quantum security architectures.	Supports the use of hybrid cryptography for transition to quantum-safe systems.
Bhargavan et al. / 2024	Formal Verification of the PQXDH Post-Quantum Key Agreement Protocol for End-to-End Secure Messaging	Formally verifies a post-quantum key agreement protocol for secure messaging.	Supports secure PQ session establishment and end-to-end communication.
Various Authors / 2024	Post-Quantum Sound CryptoVerif and Verification of Hybrid TLS and SSH Key-Exchanges	Uses formal verification to analyze hybrid PQ key-exchange protocols.	Demonstrates the importance of formally validating hybrid key-exchange security.
Various Authors / 2024	Hybrid Keys in Practice: Combining Classical, Quantum and Post-Quantum Cryptography	Studies practical methods for combining classical and post-quantum key-exchange mechanisms.	Supports our approach of combining classical X25519 with ML-KEM.
Chen et al. / 2024	GoFetch: Breaking Constant-Time Cryptographic Implementations Using Data Memory-Dependent Prefetchers	Analyzes side-channel vulnerabilities in cryptographic implementations.	Highlights the need for secure implementation of cryptographic primitives.
Various Authors / 2023	The Performance of Post-Quantum TLS 1.3	Measures PQ cryptography's effects on TLS latency, bandwidth and network performance.	Provides evidence for evaluating PQC performance and communication overhead.
Tzinos et al. / 2022	Evaluating the Performance of Post-Quantum Secure Algorithms in the TLS Protocol	Evaluates Kyber and hybrid PQ schemes under different network conditions.	Demonstrates the practical performance trade-offs of PQC.
Bürstinghaus-Steinbach et al. / 2020	Post-Quantum TLS on Embedded Systems: Integrating and Evaluating Kyber and SPHINCS+ with mbed TLS	Implements and evaluates PQ algorithms on resource-constrained embedded systems.	Supports the feasibility of PQC on constrained devices.

Table 2.1. Summary of Literature Survey

Collectively, the reviewed literature strongly supports the central design decision of this project: combining the post-quantum key-encapsulation mechanism ML-KEM-768 with the classical elliptic-curve algorithm X25519 in a hybrid construction. Studies on TLS 1.3 hybrid key agreement, PQXDH-style secure messaging, and post-quantum readiness measurements confirm that this hybrid approach is both practically deployable and increasingly

adopted across the industry. Works evaluating PQC performance in wireless, IoT, and embedded contexts further validate that ML-KEM-768 is suitable for resource-constrained mobile environments such as the Android platform targeted by this project. However, a common gap across the surveyed works is the limited exploration of combining hybrid PQ key exchange with an offline, serverless, peer-to-peer transport layer — a gap this project directly addresses.

CHAPTER 3

PROBLEM STATEMENT AND EXISTING SYSTEM

3.1 PROBLEM STATEMENT

Existing file-transfer solutions often depend on Internet/cloud infrastructure, are restricted to specific device ecosystems, or lack explicit peer identity verification and post-quantum-resistant key establishment. This creates challenges when sensitive files need to be transferred securely, particularly between nearby devices in environments where Internet connectivity is unavailable or intentionally restricted.

3.2 WHY IS IT IMPORTANT TO SOLVE?

- Sensitive data protection: Medical records, confidential documents, maps, images, and other files can be exposed if intercepted or sent to an unintended device.
- No Internet availability: Disaster zones, remote areas, and air-gapped environments may have little or no Internet connectivity.
- MITM protection: Simply discovering a nearby device does not prove that it is the intended recipient. Out-of-band pairing helps authenticate the peer before key exchange.
- Future security: RSA and ECC are vulnerable to sufficiently capable quantum computers in the near future. Using ML-KEM-768 + X25519 provides a hybrid approach combining current classical security with post-quantum protection.
- Reduced infrastructure exposure: Offline transfers require no external server or Internet connection, removing third-party infrastructure from the trust model.

3.3 EXISTING SYSTEM AND LIMITATIONS

The table below compares existing file-sharing and key-exchange solutions with the approach proposed in this project.

Solution	Weakness vs. Proposed System
Cloud drives (Drive, Dropbox)	Server-side file storage/access and dependence on Internet connectivity, versus our serverless offline transfer.
Existing Cryptosystems	Closed black-box design (e.g. DES) allows biased constants that permit designer backdoors, versus publicly reviewed post-quantum algorithms.
Traditional Nearby File Transfer	No explicit user-verified peer authentication or post-quantum key establishment, versus verified pairing combined with ML-KEM-768.
RSA / ECC key exchange	Broken by a sufficiently capable quantum computer (Shor's algorithm), versus no known existing cryptographic attacks on the proposed scheme.

Table 3.1. Comparison of Existing Systems

Our project closes the main gaps identified above: it supports air-gapped, serverless file sharing with user-verified peer authentication and uses post-quantum cryptography (ML-KEM) for key establishment.

CHAPTER 4

METHODOLOGY AND SYSTEM ARCHITECTURE

4.1 APPROACH / TECHNIQUE

- Peer Discovery: Discover and establish connections between nearby devices using the Nearby Connections API.
- Out-of-Band Pairing: Use a pairing key that is verified out-of-band by the users before key exchange.
- Hybrid Key Exchange: Use ML-KEM-768 + X25519 to establish both post-quantum and classical shared secrets for offline sessions.
- Session Key Derivation: Combine the shared secrets using HKDF-SHA256 to derive the session encryption key.
- File Encryption: Encrypt files using AES-256-GCM, providing confidentiality and integrity.
- Secure Teardown: Remove session and ephemeral private keys after the transfer.

4.2 TOOLS AND TECHNOLOGIES

- Platform: Android — Kotlin
- UI: Jetpack Compose + Material 3
- Cryptography: Bouncy Castle
- Cryptographic Components: ML-KEM-768, X25519, HKDF-SHA256, AES-256-GCM
- Offline Connectivity: Nearby Connections API
- Pairing: Authentication token

4.3 SYSTEM ARCHITECTURE — OFFLINE MODE

In the offline architecture, two peer devices (referred to as Alice and Bob) discover each other on the local network, perform out-of-band pairing verification, and then carry out a post-quantum hybrid key exchange before any file data is transferred. The sequence of steps is as follows:

1. Discover peer: Each device finds the other on the local network using the Nearby Connections API.

2. Out-of-band pairing: One device displays a QR code (or pairing number) which encodes a pairing value; the other device scans or enters the code to confirm the same pairing number.
3. Post-quantum key exchange: Devices exchange public keys (X25519 + ML-KEM-768); the initiating device encapsulates a secret using the peer's ML-KEM public key.
4. Decapsulation: The receiving device decapsulates the secret and both sides now hold the KEM ciphertext and ECDH key material.
5. Derive shared secret: HKDF-SHA256 is applied over both the ML-KEM and X25519 secrets to derive a single session key.
6. Encrypted file transfer: The file is encrypted and transferred using AES-256-GCM over the established channel.
7. Discard session key: Ephemeral keys and the session key are wiped from memory once the transfer completes.

4.4 DATA FLOW DIAGRAM (LEVEL 1)

Figure 4.1 presents the Level 1 Data Flow Diagram (DFD) of the Secure File Sharing Application. The entire system operates inside an offline, air-gapped boundary, and interacts with two external entities: the Sender (Alice) and the Receiver (Bob). Two data stores are used: D1 (File Storage on the local filesystem) and D2 (Keystore held in ephemeral RAM). The system is decomposed into six processes:

- 1.0 Nearby Discovery and OOB Verification: Receives the discovery request and token from the sender and the accept/QR-scan response from the receiver, and produces an authenticated session event.
- 2.0 Hybrid Key Exchange (ML-KEM-768 + X25519): Exchanges KEM public keys and ciphertext with the receiver and stores the shared post-quantum and ECDH secrets in the keystore.
- 3.0 Session Key Derivation (HKDF-SHA256): Retrieves the raw secrets from the keystore and writes back the derived 256-bit session key.
- 4.0 Authenticated Encryption (AES-256-GCM): Reads the plaintext file from D1 and the session key from D2, and outputs the ciphertext together with the IV and authentication tag.
- 5.0 Authenticated Decryption and Tag Verification: Verifies the authentication tag, and writes the decrypted plaintext to D1 only if verification succeeds.
- 6.0 Secure Key Teardown: On transfer completion, wipes the session key and ephemeral private keys from the keystore.

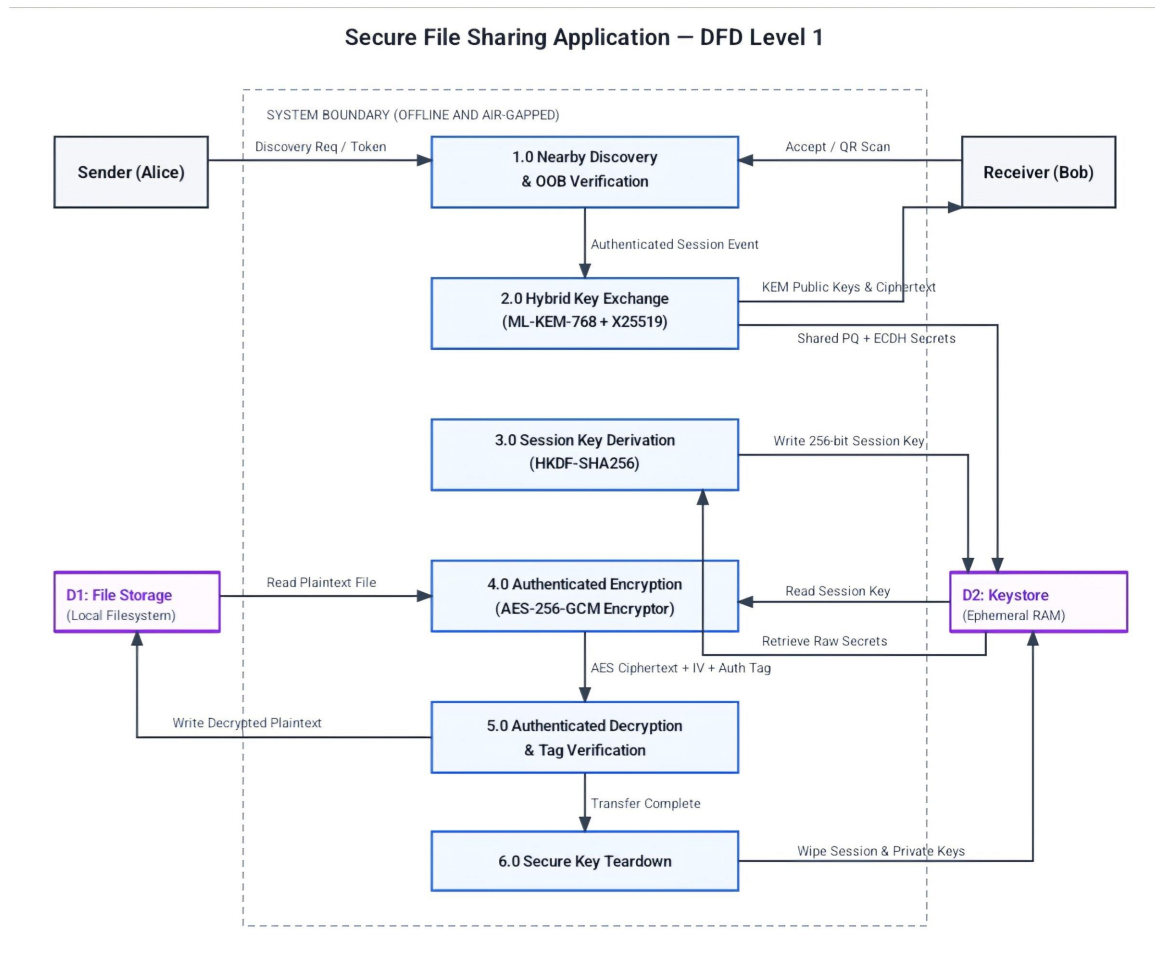


Figure 4.1. Data Flow Diagram (Level 1) of the Secure File Sharing Application

4.5 ACTIVITY DIAGRAM

Figure 4.2 shows the activity diagram of the offline peer-to-peer transfer, which is organised into five phases. In Phase 1, the devices advertise, discover each other and request a connection through the Nearby Connections API. In Phase 2, an authentication code or QR code is displayed on one device and compared out-of-band on the other; if the codes do not match, the connection is rejected and the devices disconnect.

In Phase 3, the devices exchange ephemeral ML-KEM-768 and X25519 public keys, the sender encapsulates a secret that the receiver decapsulates, and the final symmetric key is derived using HKDF-SHA256 over the KEM and ECDH secrets. In Phase 4, the file is encrypted with AES-256-GCM and transmitted together with its authentication tag; the receiver decrypts and saves the file only if the tag is valid, otherwise an integrity error is raised. Phase 5 purges the ephemeral private keys and derived session keys from RAM, on both the successful and the failed paths.

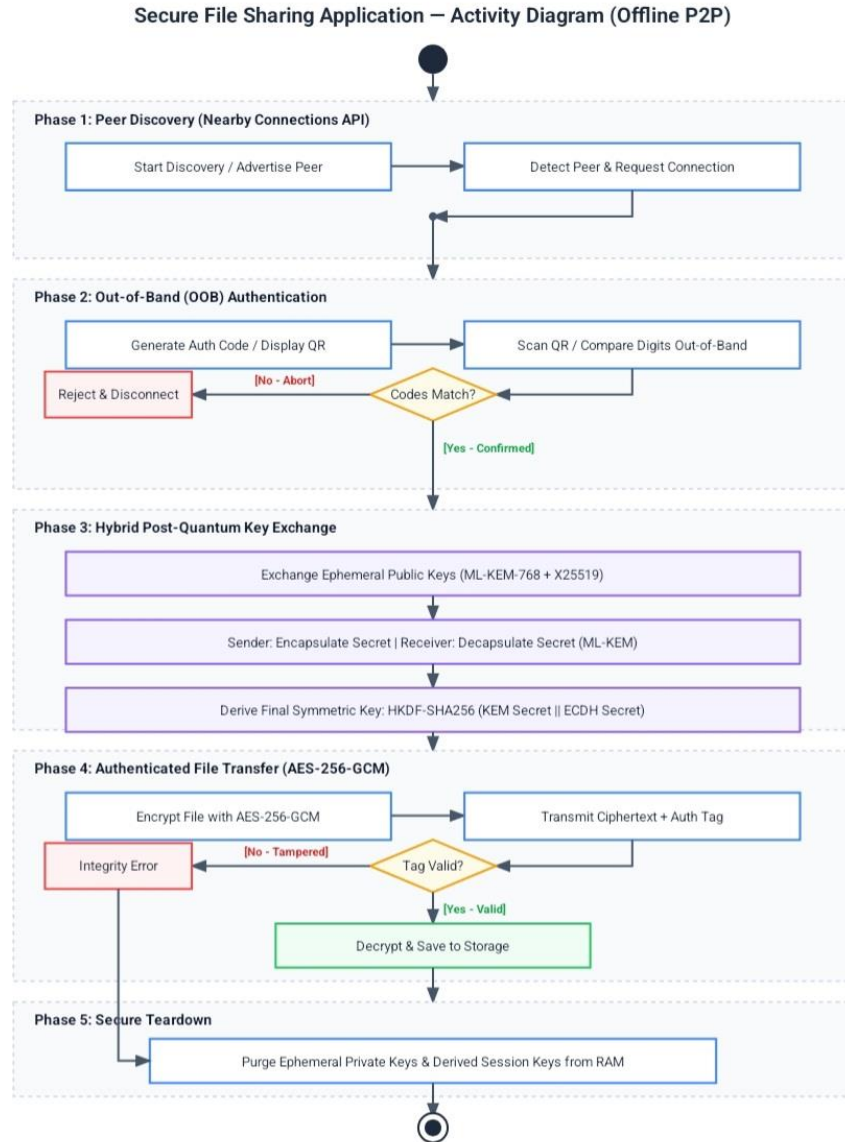


Figure 4.2. Activity Diagram of the Offline Peer-to-Peer File Transfer

4.6 END-TO-END USE CASE

The table below traces the complete end-to-end use case for the offline mode, along with the implementation status of each step at the time of this review.

Use Case Step	Module(s) Involved	Status	Notes
1. Device Discovery	Nearby Connections API	Live	Device discovery happens over multiple channels on supported devices.
2. Out-of-band pairing	Pairing (magic number)	Live	Number generated; verification flow stubbed.
3. Key Exchange	Quantum-Secure KEX	Live	Keys generated and exchanged in-memory; not yet over network.

Use Case Step	Module(s) Involved	Status	Notes
4. Derive session key	HKDF-SHA256 Key Gen	Live	Combines KEM + ECDH secrets; runs on both sides.
5. Send encrypted file	Transfer (AES-256-GCM)	Live	Encrypt/decrypt with auth-tag verification; file I/O stubbed.

Table 4.1. End-to-End Use Case Status

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

At the time of this first review, the core cryptographic and connectivity building blocks of the Secure File Sharing Application have been implemented and individually verified, while full end-to-end wiring of the offline data channel remains ongoing. The following results have been achieved:

- Successfully established offline peer-to-peer device discovery and connection using the Nearby Connections API without requiring Internet connectivity.
- Implemented user-verified device authentication using authentication digits before accepting the connection.
- Implemented the ML-KEM-768 + X25519 hybrid key-exchange workflow for establishing session secrets.
- Implemented HKDF-SHA256 to derive the final session encryption key from the established shared secrets.
- Implemented AES-256-GCM encryption and decryption, including authentication-tag verification for file integrity.
- Successfully verified encrypted file transfer and authenticated decryption between connected devices.
- Ensured that plaintext file data is not transmitted during the encrypted transfer process.
- Implemented session-specific ephemeral keys, preventing reuse of long-term secrets for individual transfers.
- Implemented secure session handling by removing ephemeral keys and session keys after transfer.
- Designed the system to minimize dependence on external infrastructure for offline file sharing.
- Demonstrated the feasibility of combining traditional networking, post-quantum cryptography, and authenticated encryption in a mobile file-sharing application.

5.2 APPLICATION SCREENSHOTS AND SESSION LOG

Figures 5.1 and 5.2 show the working Android application during an offline transfer between two devices. The Nearby tab (Figure 5.1) displays the local device ID (D7BC08AE) and the connected peer, Pixel 7a (device ID 172A2635), along with the Send File and

Disconnect controls. A confirmation line above the connection card reports that the file "IMG-20260930-WA0000.jpg" (66960 bytes) was received.

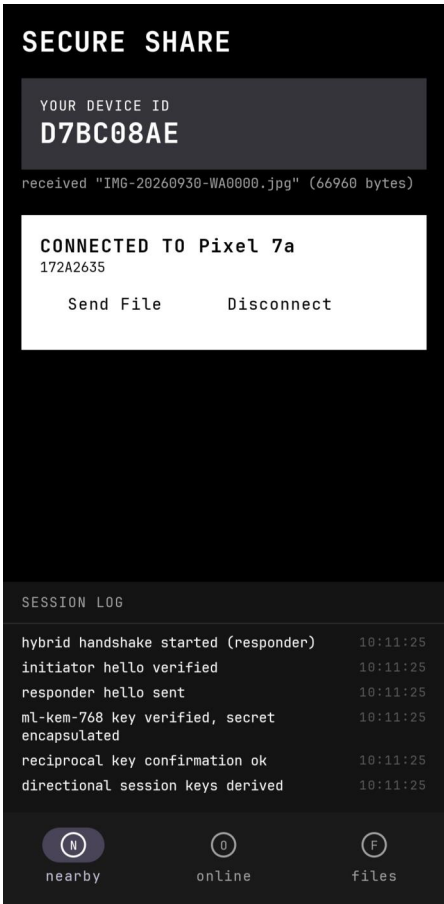


Figure 5.1. Nearby Tab: Connected Peer and Session Log



Figure 5.2. Received Files Tab

The session log at the bottom of Figure 5.1 records the hybrid handshake performed before the file was accepted. The logged events are summarised in Table 5.1. All six events carry the same timestamp (10:11:25), indicating that the complete handshake, including the ML-KEM-768 encapsulation and key derivation, finished within a second.

Step	Session Log Entry	Meaning
1	hybrid handshake started (responder)	The hybrid key-exchange handshake begins, with this device acting as the responder.
2	initiator hello verified	The initiating peer's hello message is received and successfully verified.
3	responder hello sent	This device replies with its own hello message.
4	ml-kem-768 key verified, secret encapsulated	The peer's ML-KEM-768 public key is verified and a shared secret is encapsulated.
5	reciprocal key confirmation ok	Both peers confirm that they hold matching key material.
6	directional session keys derived	Separate session keys are derived for each direction of transfer.

Table 5.1. Handshake Events Recorded in the Session Log

The Received Files tab (Figure 5.2) lists the transferred file with its name, size (65.4 KB), transfer mode (nearby), sender (Pixel 7a) and the time of receipt (2026-09-30 10:11), together with open and save options. The reported size matches the 66960 bytes shown in the Nearby tab, confirming that the file was received intact.

5.3 DISCUSSION

The results obtained so far validate the core design premise of the project — that a hybrid post-quantum key-exchange scheme can be practically implemented on a resource-constrained Android device without introducing prohibitive overhead. The offline pathway (Nearby Connections + out-of-band pairing + ML-KEM-768/X25519 + HKDF-SHA256 + AES-256-GCM) has been verified module-by-module, with cryptographic primitives producing correct, authenticated ciphertext, and the pairing verification correctly preventing unauthenticated peers from proceeding to key exchange.

The main outstanding item at this stage is the direct integration of the negotiated session key with the live Nearby Connections data channel — currently, key exchange has been verified in-memory rather than over the actual network transport, and file I/O for the offline pathway remains stubbed. These integration steps, along with QR-based pairing and performance benchmarking, are planned for the next phase of the project.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

This project demonstrates a secure offline peer-to-peer file-sharing architecture. The system combines Nearby Connections, user-verified pairing, ML-KEM-768 + X25519, HKDF-SHA256, and AES-256-GCM to provide secure local file transfer without Internet or server dependency. The Nearby Connections API is specifically designed for nearby offline data exchange and supports user verification of connection authentication codes, making it well suited to this use case.

Overall, the proposed architecture demonstrates the feasibility of combining offline peer-to-peer communication, post-quantum cryptography, and authenticated encryption into a single, secure file-sharing system. Extending the system to online, multi-recipient sharing is planned for Phase II of the project.

6.2 FUTURE ENHANCEMENTS

- QR-based pairing: Add QR-code verification as an alternative to manually comparing authentication digits.
- Digital signatures: Add ML-DSA-based device identity verification for stronger authentication.
- Large-file optimization: Introduce chunked encryption and resumable transfers for large files.
- Multi-device support: Extend offline sharing to controlled multi-device sessions.
- Online multi-recipient sharing (Phase II): Add a ciphertext server so files are encrypted with a random symmetric key that is distributed to each recipient using ML-KEM-768.
- Performance evaluation: Benchmark encryption, key exchange, transfer speed, memory usage, and battery consumption on different Android devices.
- Secure identity management: Add persistent device identities and trusted-device management.
- Security testing: Perform penetration testing and formal security analysis of the complete protocol.

CHAPTER 7

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